CSE 590 Advanced Programming Languages
Project Proposal

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Due on CANVAS Thursday, October 8th, 2026, **11:59pm**

1 The Project

The goals of the project component in this course include:

- To develop a familiarity with the existing literature in your field.
- To give you a setting in which to practice computer science research on a compressed timeline.
- To connect your own intellectual interests with Programming Languages research.

If you are already working on a research project, we highly encourage you to propose something related to it, as long as you can make some kind of PL connection.

This project proposal is the first stage of our compressed research timeline. In research, we have to decide what projects to work on and often we need to apply for research funding (grants, fellowships, awards). Writing the proposal gives you a chance to consider why you should do the research you propose. When writing, consider the following questions:

- What is the problem with the current state of the art that you are trying to address? If you are not sure, then identify prior works you need to familiarize yourself with.
- What are specific, testable hypotheses that you have about your proposed work?
- How do you plan to evaluate whether your project succeeded?
- In the best case scenario where your research project completely succeeded, in what way would you have improved the state of your field?

Since many members of the class have not conducted research before, we don't expect you to have strong answers to all of these questions, but we do expect you to think about them thoroughly and try your best.

Over the course of the semester, you may encounter setbacks or find more promising approaches than what you originally considered. This is completely normal and expected. The point of the project proposal is not to say exactly what you will do in the project, but to be a starting point for your rigorous intellectual pursuit.

2 Timeline and Deliverables

There are two deadlines to keep in mind after your proposal: a midterm update is due on Thursday, November 5th, 2026 and the final project submission will be due on Thursday, December 10th, 2026, both at 11:59pm on Canvas.

The midterm update will be a report on how the project is going and what deviations if any from the original proposal need to be made. Your final project submission will consist of a research report (10–20 pages) as well as submitting any associated additional artifacts (software, appendices).

3 Format

Submit your proposal as a pdf written in the PACMPL format (see the template on Canvas). The proposal should include the following sections:

- Title and authors
- Abstract: a brief 1-2 paragraph summary of the project.
- Background: your current understanding of the state of the art and motivation for your proposal. Include references to background work you have read and/or plan to read.
- Research Plan: what your specific hypothesis is and how you plan to evaluate it over the course of the semester.
- Deliverables: an overview of what you plan to include in your final project report and associated research artifacts (i.e., code, detailed proofs etc).
- Timeline: a rough estimate of when you will complete each component of the project. Take into consideration what you want to have completed by the midterm update and the final deadline.
- Collaboration and AI disclosure: List any collaborators you have outside of the course, and what if any generative AI tools were used in developing this proposal. Additionally, if your project relates to generative AI tools, include a description of how you will use them, consistent with the guidelines in the syllabus.
- References: these should be made using bibtex.

The proposal should be roughly 2–4 pages, not counting the references.